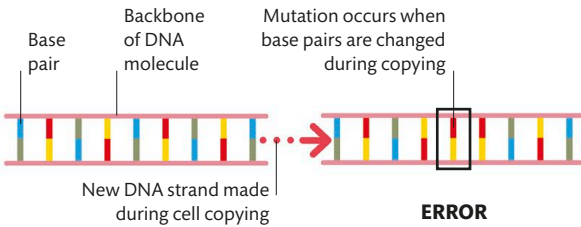




MUTATION

When cells divide, the double-stranded DNA splits into single strands, and each base is matched with a new complementary base to form two new copies of the DNA. However, sometimes copying produces changes in the sequence. These may cause a gene to produce an altered protein or stop it from working at all. Mutations may arise during life or may be inherited from parents.



**AT LEAST ONE-THIRD
OF ALL OUR GENES
ARE ACTIVE PRIMARILY
IN THE BRAIN**

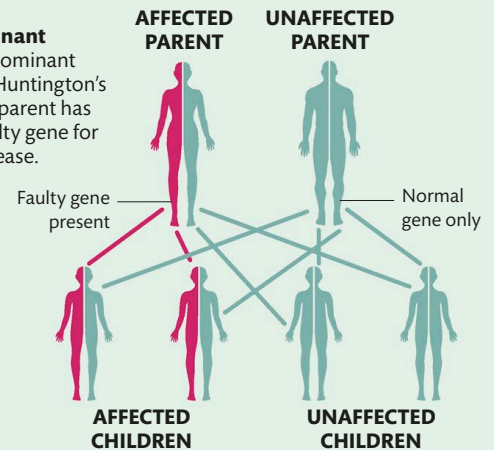


How faulty genes affect the brain

Genes do not directly control behavior; instead, they govern the number and characteristics of nerve cells whose actions combine to produce our mental functions. For example, some genes influence the levels of neurotransmitters (see p.24), which in turn regulate functions such as memory, mood, behavior, and cognitive skills. A faulty gene may fail to produce a protein needed for healthy brain function or may increase the risk of a disorder such as Alzheimer's disease. Some faults can be inherited from parents; two inheritance patterns are shown here.

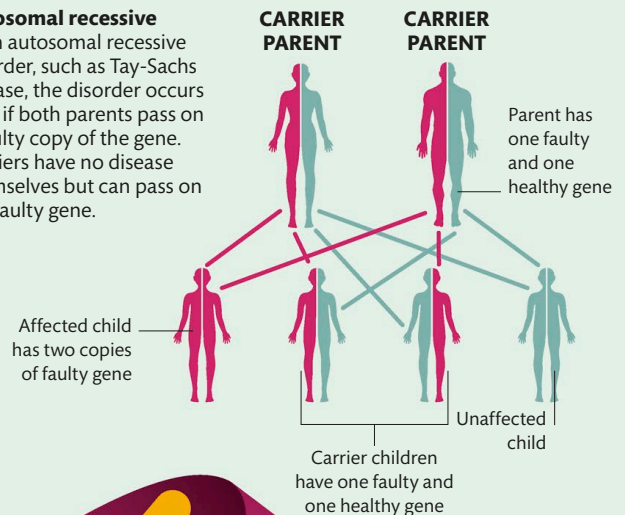
Autosomal dominant

In an autosomal dominant disorder, such as Huntington's disease, only one parent has to pass on the faulty gene for it to cause the disease.



Autosomal recessive

In an autosomal recessive disorder, such as Tay-Sachs disease, the disorder occurs only if both parents pass on a faulty copy of the gene. Carriers have no disease themselves but can pass on the faulty gene.



Guanine (blue) always bonds with cytosine (green)